Single-cell RNA-seq — Cell-type Report

Sample deliverable · 10x pbmc3k

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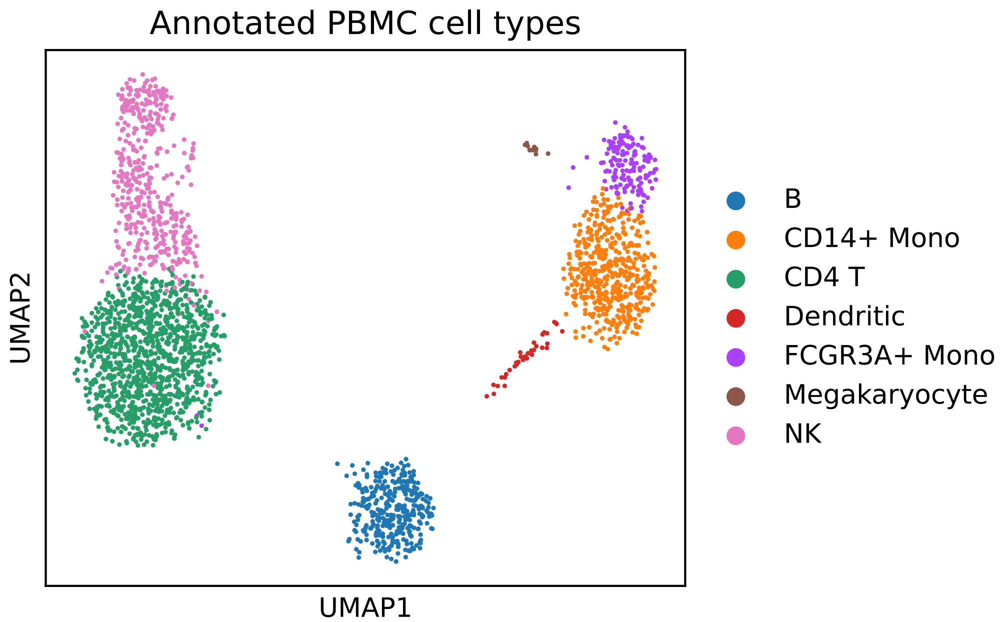
Executive summary. A raw 10x single-cell matrix was taken end-to-end — quality control, normalisation, clustering and **marker-based cell-type annotation** — to a labelled map of the immune populations present. **2,638 quality-controlled cells** resolved into **7 annotated cell types**, each validated by canonical markers.

Cell-type composition

cell_type	n_cells	pct
CD4 T	1138	43.1
CD14+ Mono	486	18.4
NK	474	18.0
B	341	12.9
FCGR3A+ Mono	150	5.7
Dendritic	36	1.4
Megakaryocyte	13	0.5



What this means. The clusters are *biologically interpretable* — T cells, monocyte subsets, B, NK, dendritic and a small megakaryocyte population — not merely statistically distinct. That internal consistency is the real quality signal.

Caveats. Cluster resolution is a question-driven choice; CD8 T cells sit adjacent to CD4 T at this depth and are honestly flagged rather than forced apart. Multi-sample work additionally needs doublet detection and batch integration before cross-condition comparison.

Reproducibility. The pipeline re-runs from a clean clone in minutes and is verified by continuous integration. Code and live report: `github.com/pgristow/scrnaseq-pbmc-scanpy`.

Prepared by Peter Ristow, PhD — Roche Principal Scientist (NGS). Production engagements use a private, hardened pipeline on your data under NDA.